MTH 201 Improper Integrals

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. For what positive numbers a does $\int_1^\infty a^x\,dx$ converge?

$$\int_1^\infty a^x\,dx = \lim_{t\to\infty}\left[\int_1^t a^x\,dx\right] = \lim_{t\to\infty}\left[\left.\frac{a^x}{\ln(a)}\right|_1^t\right] = \lim_{t\to\infty}\left[\frac{a^t}{\ln(a)} - \frac{a}{\ln(a)}\right] = \lim_{t\to\infty}\frac{a^t}{\ln(a)} - \lim_{t\to\infty}\frac{a}{\ln(a)}.$$

We see that the second limit here is just a constant, so as long as the first limit evaluates to a finite value the integral converges. So, we need to determine for what positive values a the limit $\lim\limits_{t\to\infty} a^t$ evaluates to be a finite number.

This will happen if $0 < a \le 1$ because if $a = 1$ then $\lim\limits_{t\to\infty} 1^t = 1$ and if $0 < a < 1$ then $\lim\limits_{t\to\infty} a^t = 0$.

Therefore, the integral $\int_1^\infty a^x\,dx$ converges when $0 < a \le 1$.

2. Consider each of the following questions.

a) Show that $\int_{-\infty}^\infty x\,dx$ is divergent.

First we write the integral as

$$\int_{-\infty}^\infty x\,dx = \int_{-\infty}^0 x\,dx + \int_0^\infty x\,dx = \lim_{t\to-\infty}\left[\int_t^0 x dx\right] + \lim_{s\to\infty}\left[\int_0^s x dx\right] = \lim_{t\to-\infty}\left[\left.\frac{1}{2}x^2\right|_t^0\right] + \lim_{s\to\infty}\left[\left.\frac{1}{2}x^2\right|_0^s\right]$$

$$= \lim_{t\to-\infty}\left[-\frac{1}{2}t^2\right] + \lim_{s\to\infty}\left[\frac{1}{2}s^2\right] = -\infty + \infty$$

Since both parts of this integral diverge the overall integral diverges.

b) Show that $\lim\limits_{t\to\infty}\left[\int_{-t}^t x\,dx\right] = 0$

$$\lim_{t\to\infty}\left[\int_{-t}^t x\,dx\right] = \lim_{t\to\infty}\left[\left.\frac{1}{2}x^2\right|_{-t}^t\right] = \lim_{t\to\infty}\left[\frac{1}{2}t^2 - \frac{1}{2}t^2\right] = \lim_{t\to\infty}[0] = 0$$

c) What do parts (a) and (b) show us?

This shows us that we cannot say that $\int_{-\infty}^\infty x\,dx = \lim\limits_{t\to\infty}\left[\int_{-t}^t x\,dx\right]$

3. If $\int_{-\infty}^{\infty} f(x)\,dx$ is convergent, *a* and *b* are real numbers, and $f(x)$ is continuous with an antiderivative of $F(x)$, show that $\int_{-\infty}^{a} f(x)\,dx+\int_{a}^{\infty} f(x)\,dx=\int_{-\infty}^{b} f(x)\,dx+\int_{b}^{\infty} f(x)\,dx$ regardless of the values of *a* and *b*.

We see that

$$\begin{aligned}\int_{-\infty}^{a} f(x)\,dx+\int_{a}^{\infty} f(x)\,dx &= \lim_{t\to-\infty}\left[\int_{t}^{a} f(x)\,dx\right]+\lim_{s\to\infty}\left[\int_{a}^{s} f(x)\,dx\right]=\lim_{t\to-\infty}\left[F(a)-F(t)\right]+\lim_{s\to\infty}\left[F(s)-F(a)\right]\\ &=F(a)-\lim_{t\to-\infty}F(t)+\lim_{s\to\infty}F(s)-F(a)=\lim_{s\to\infty}F(s)-\lim_{t\to-\infty}F(t)\\ \int_{-\infty}^{b} f(x)\,dx+\int_{b}^{\infty} f(x)\,dx &= \lim_{t\to-\infty}\left[\int_{t}^{b} f(x)\,dx\right]+\lim_{s\to\infty}\left[\int_{b}^{s} f(x)\,dx\right]=\lim_{t\to-\infty}\left[F(b)-F(t)\right]+\lim_{s\to\infty}\left[F(s)-F(b)\right]\\ &=F(b)-\lim_{t\to-\infty}F(t)+\lim_{s\to\infty}F(s)-F(b)=\lim_{s\to\infty}F(s)-\lim_{t\to-\infty}F(t)\end{aligned}$$

Clearly both sets of integral simplify to $\lim_{s\to\infty}F(s)-\lim_{t\to-\infty}F(t)$ and are therefore equal.

4. An important concept in Engineering is the *Laplace transform* of *f*, which is the function *F* defined by $s>0$ by

$$F(s)=\int_{0}^{\infty} f(t)e^{-st}\,dt$$

a) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t)=1$.

$$\begin{aligned}F(s)&=\int_{0}^{\infty} f(t)e^{-st}\,dt=\int_{0}^{\infty} e^{-st}\,dt=\lim_{b\to\infty}\left[\int_{0}^{b} e^{-st}\,dt\right]=\lim_{b\to\infty}\left[-\frac{1}{s}e^{-st}\Big|_{0}^{b}\right]\\ &=\lim_{b\to\infty}\left[-\frac{1}{s}e^{-sb}+\frac{1}{s}e^{0}\right]=\lim_{b\to\infty}\left[-\frac{1}{se^{sb}}+\frac{1}{s}\right]=0+\frac{1}{s}=\frac{1}{s}\end{aligned}$$

b) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t)=t$.

$$F(s)=\int_0^\infty f(t)e^{-st}dt=\int_0^\infty te^{-st}dt=\lim_{b\to\infty}\left[\int_0^b te^{-st}dt\right]$$

Using tabular integration by parts we have that

t	e^{-st}	Skip
1	$-\frac{1}{s}e^{-st}$	+
0	$\frac{1}{s^2}e^{-st}$	-

$$\lim_{b\to\infty}\left[\int_0^b te^{-st}dt\right]=\lim_{b\to\infty}\left[-\frac{t}{s}e^{-st}-\frac{1}{s^2}e^{-st}\Big|_0^b\right]=\lim_{b\to\infty}\left[\left(-\frac{b}{s}e^{-sb}-\frac{1}{s^2}e^{-sb}\right)-\left(0-\frac{1}{s^2}e^0\right)\right]$$

$$=\lim_{b\to\infty}\left[\left(-\frac{b}{se^{sb}}-\frac{1}{s^2e^{sb}}\right)+\frac{1}{s^2}\right]=\lim_{b\to\infty}\left[-\frac{b}{se^{sb}}\right]+\lim_{b\to\infty}\left[-\frac{1}{s^2e^{sb}}+\frac{1}{s^2}\right]$$

$$\overset{H}{=}\lim_{b\to\infty}\left[-\frac{1}{s^2e^{sb}}\right]+\left[0+\frac{1}{s^2}\right]$$

$$=[0]+\left[\frac{1}{s^2}\right]$$

$$=\frac{1}{s^2}$$

c) Find the Laplace transform (that is find a simplified form of $F(s)$) when $f(t)=e^{at}$ (where a is just some real number).

$$F(s)=\int_0^\infty f(t)e^{-st}dt=\int_0^\infty e^{at}e^{-st}dt=\int_0^\infty e^{(a-s)t}dt$$

Note that if $s=a$ then $F(s)=\int_0^\infty e^{(a-s)t}dt=\int_0^\infty e^0dt=\int_0^\infty dt=\lim_{b\to\infty}\left[\int_0^b dt\right]=\lim_{b\to\infty}[b-0]=\infty$ and therefore diverges.

If $s\neq a$ then there are two cases to consider.

If $s<a$ then

$$F(s)=\int_0^\infty e^{(a-s)t}dt=\lim_{b\to\infty}\left[\int_0^b e^{(a-s)t}dt\right]=\lim_{b\to\infty}\left[\frac{1}{a-s}e^{(a-s)t}\Big|_0^b\right]$$

$$=\lim_{b\to\infty}\left[\frac{1}{a-s}e^{(a-s)b}-\frac{1}{a-s}e^0\right]=\lim_{b\to\infty}\left[\frac{1}{a-s}e^{(a-s)b}+\frac{1}{a-s}\right]=\infty+\frac{1}{a-s}=\infty$$

If $s>a$ then

$$F(s)=\int_0^\infty e^{(a-s)t}dt=\lim_{b\to\infty}\left[\int_0^b e^{(a-s)t}dt\right]=\lim_{b\to\infty}\left[\frac{1}{a-s}e^{(a-s)t}\Big|_0^b\right]$$

$$=\lim_{b\to\infty}\left[\frac{1}{a-s}e^{(a-s)b}-\frac{1}{a-s}e^0\right]=\lim_{b\to\infty}\left[\frac{1}{a-se^{(s-a)b}}-\frac{1}{a-s}\right]=0-\frac{1}{a-s}=\frac{1}{s-a}$$

5. Show how the Comparison Theorem can be used to help determine if the following integrals converge or diverge.

a) $\displaystyle\int_1^\infty \frac{1}{\left[x+\ln(x)\right]^2}\,dx$

Note that for values of x on $[1,\infty)$ we have that $\dfrac{1}{\left[x+\ln(x)\right]^2} \le \dfrac{1}{x^2}$.

We see that $\displaystyle\int_1^\infty \frac{1}{x^2}\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{1}{x^2}\,dx\right] = \lim_{t\to\infty}\left[-\frac{1}{x}\Big|_1^t\right] = \lim_{t\to\infty}\left[-\frac{1}{t}+1\right] = 1$.

Since $\displaystyle\int_1^\infty \frac{1}{x^2}\,dx$ converges, by the Comparison Theorem, $\displaystyle\int_1^\infty \frac{1}{\left[x+\ln(x)\right]^2}\,dx$ must also converge.

b) $\displaystyle\int_1^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx$

Note that for values of x on $[1,\infty)$ we have that $\dfrac{x+1}{\sqrt{x^4-x}} \ge \dfrac{x+1}{x^4} \ge \dfrac{1}{x^4}$.

$$\int_1^\infty \frac{1}{x^4}\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{1}{x^4}\,dx\right]$$
$$= \lim_{t\to\infty}\left[\frac{-1}{3x^3}\Big|_1^t\right] = \lim_{t\to\infty}\left[-\frac{1}{3t^3}+\frac{1}{3(1)^2}\right]$$
$$= 0+\frac{1}{3} = \frac{1}{3}$$

Since $\displaystyle\int_1^\infty \frac{1}{x^4}\,dx$ converges, by the Comparison Theorem, $\displaystyle\int_1^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx$ must also converge.

c) $\displaystyle\int_0^\infty \frac{x}{x^4+1}\,dx$

Note that for values of x on $[1,\infty)$ we have that $\dfrac{x}{x^4+1} \le \dfrac{x}{x^4} = \dfrac{1}{x^3}$.

$$\int_1^\infty \frac{1}{x^3}\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{1}{x^3}\,dx\right]$$
$$= \lim_{t\to\infty}\left[\frac{-1}{2x^2}\Big|_1^t\right] = \lim_{t\to\infty}\left[-\frac{1}{2t^2} + \frac{1}{2(1)^2}\right]$$
$$= 0 + \frac{1}{2} = \frac{1}{2}$$

Since $\displaystyle\int_1^\infty \frac{1}{x^3}\,dx$ converges, by the Comparison Theorem, $\displaystyle\int_1^\infty \frac{x}{x^4+1}\,dx$ must also converge. However note that since $\displaystyle\int_0^1 \frac{x}{x^4+1}\,dx$ must be finite that $\displaystyle\int_0^\infty \frac{x}{x^4+1}\,dx = \int_0^1 \frac{x}{x^4+1}\,dx + \int_1^\infty \frac{x}{x^4+1}\,dx$ must also converge.

d) $\displaystyle\int_0^1 \frac{\sec^2(x)}{x\sqrt{x}}\,dx$

Note that for values of x on $(0,1]$ we have that $\dfrac{\sec^2(x)}{x\sqrt{x}} \ge \dfrac{1}{x^{3/2}}$.

$$\int_0^1 \frac{1}{x^{3/2}}\,dx = \lim_{t\to 0^+}\left[\int_t^1 \frac{1}{x^{3/2}}\,dx\right]$$
$$= \lim_{t\to 0^+}\left[-2x^{-1/2}\Big|_t^1\right] = \lim_{t\to 0^+}\left[-2(1)^{-1/2} + 2(t)^{-1/2}\right]$$
$$= \lim_{t\to 0^+}\left[-2 + \frac{2}{\sqrt{t}}\right] = -2 + \infty = \infty$$

Since $\displaystyle\int_0^1 \frac{1}{x^{3/2}}\,dx$ diverges, by the Comparison Theorem, $\displaystyle\int_0^1 \frac{\sec^2(x)}{x\sqrt{x}}\,dx$ must also diverge.

e) $\displaystyle\int_3^{\infty} \frac{x^3 - x + 2}{x^5 + x^4 - x^3}\,dx$

Note that for values of x on $[3,\infty)$ we have that $\displaystyle\frac{x^3 - x + 2}{x^5 + x^4 - x^3} \le \frac{x^3}{x^5 + x^4 - x^3} \le \frac{x^3}{x^5 - x^3} = \frac{x^3}{x^3\left(x^2 - 1\right)} = \frac{1}{x^2 - 1}$

We see that

$$\begin{aligned}
\int_3^{\infty} \frac{1}{x^2-1}\,dx &= \lim_{t\to\infty}\left[\int_3^{t} \frac{1}{x^2-1}\,dx\right] = \lim_{t\to\infty}\left[\int_3^{t} \frac{1}{(x-1)(x+1)}\,dx\right] \\
&= \lim_{t\to\infty}\left[\int_3^{t}\left[\frac{1}{2(x-1)} - \frac{1}{2(x+1)}\right]dx\right] = \lim_{t\to\infty}\left[\left(\frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1|\right)\Bigg|_3^t\right] \\
&= \lim_{t\to\infty}\left[\left(\frac{1}{2}\ln\left(\frac{x-1}{x+1}\right)\right)\Bigg|_3^t\right] = \lim_{t\to\infty}\left[\frac{1}{2}\ln\left(\frac{t-1}{t+1}\right) - \frac{1}{2}\ln\left(\frac{2}{4}\right)\right] \overset{H}{=} \lim_{t\to\infty}\left[\frac{1}{2}\ln(1) - \frac{1}{2}\ln\left(\frac{1}{2}\right)\right] \\
&= 0 - \frac{1}{2}\ln\left(\frac{1}{2}\right) = \frac{1}{2}\ln(2)
\end{aligned}$$

Since $\displaystyle\int_3^{\infty} \frac{1}{x^2-1}\,dx$ converges, by the Comparison Theorem, $\displaystyle\int_3^{\infty} \frac{x^3 - x + 2}{x^5 + x^4 - x^3}\,dx$ must also converge.

f) $\displaystyle\int_3^{\infty} \frac{1}{\cos(x) - 2 + \ln(x)}\,dx$

Note that for values of x on $[3,\infty)$ we have that

$$\frac{1}{\cos(x) - 2 + \ln(x)} \ge \frac{1}{1 - 2 + \ln(x)} = \frac{1}{-1 + \ln(x)} \ge \frac{1}{\ln(x)} \ge \frac{1}{x}$$

We see that $\displaystyle\int_3^{\infty} \frac{1}{x}\,dx = \lim_{t\to\infty}\left[\int_3^{t} x\,dx\right] = \lim_{t\to\infty}\left[\frac{1}{2}x^2\Bigg|_3^t\right] = \lim_{t\to\infty}\left[\frac{1}{2}t^2 - \frac{9}{2}\right] = \infty$

Since $\displaystyle\int_3^{\infty} \frac{1}{x}\,dx$ diverges, by the Comparison Theorem, $\displaystyle\int_3^{\infty} \frac{1}{\cos(x) - 2 + \ln(x)}\,dx$ must also diverge.

6. Suppose Kyle is trying to make the following comparison

$$\frac{x^3 - x + 2}{x^5 + x^4 - x^3} \leq \frac{x^3}{x^5 + x^4 - x^3} \leq \frac{x^3}{-x^3} = -1$$ where x represents values on the interval $\left[2,\infty\right)$.

Is this comparison valid? Why or why not?

The first step of this comparison is valid because on the interval $\left[2,\infty\right)$ the expression $\left(-x+2\right)$ is either 0 or negative and so the numerator of the fraction is unchanged or made larger by the removal of this term.

The second step of this comparison is NOT valid. While $\left(x^5 + x^4\right)$ being removed means that the denominator is made smaller it is also made negative which makes the entire fraction $\frac{x^3}{-x^3}$ represent a negative number while $\frac{x^3}{x^5 + x^4 - x^3}$ represents a positive number. Therefore, inequality here is facing the wrong direction.